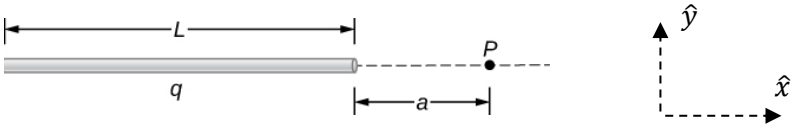
Physics 2 Problem Solving – Chapter 39 More on the Electric Potential

1) A Thin wire of length L with a uniform linear charge density of λ is shown below, determine the electric potential (V) along the x-axis at point P due to the charge on the wire. Draw and label on the diagram the incremental charge, radius, angle and any other variables needed to solve the problem.



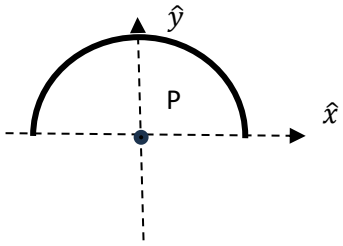
$Answer$: $V(r) = k\lambda\left[\frac{(L+a)}{a}\right]$

2) Shown below is a semicircular thin wire with radius R and a uniform linear charge density of λ.

A) Draw and label on the diagram the incremental charge, radius, angle, and any other variables needed to solve the problem.

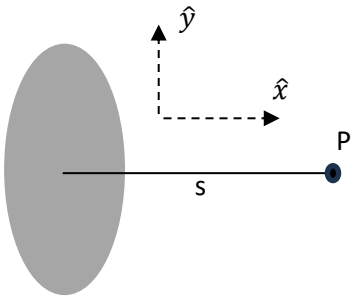
B) Determine the electric potential (V) at point P which is at the radius of the semicircle.

$Answer$: $V(r) = k\lambda\pi$



3) A circular disk with a radius R and a uniform surface charge density of σ is shown below, determine the electric potential at point P which is at a distance of s from the center of the disk. Draw and label on the diagram the incremental charge, radius, angle, and any other variables needed to solve the problem.

Answer: $V(r) = k\sigma 2\pi\left[\sqrt{R^2 + s^2} \ - s\right]$



4) A positively charged straight wire with a uniform linear charge density of λ, determine the electric field strength at point P, y meters above the wire and situated above the end of the wire of length L. Assume a and L are constant and do not change in this problem.

Answer: $V(r) = k\lambda\left[\ln\left(L + \sqrt{L^2 + a^2}\right) - \ln(a)\right]$

